

Scalability & Future Plan

Date	1 October 2026
Team ID	SWTID-2026-1235
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	No user accounts or learning history	Learners cannot save or revisit previous sessions	High
2	Runs locally only	Application is not accessible to multiple users remotely	Medium
3	Dependence on Gemini API quota	Heavy usage may cause delays or API limitations	Medium
4	No multilingual or voice support	Limits accessibility for users who prefer regional languages or voice interaction	Low
5	No learning analytics	Users cannot track their learning progress over time	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local server	Containerise with Docker and deploy on a cloud platform with load balancing
2	Data Storage	Stateless	Add PostgreSQL/Firestore for user accounts and learning history
3	Performance	Direct API calls	Add response caching and asynchronous request handling
4	Security	API key stored in .env	Add authentication, rate limiting and cloud secret management
5	Monitoring	Basic local testing	Add application monitoring, logging and performance monitoring

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User login and saved learning history	1–2 months	Personalised continuity
Phase 3	Multi-language and voice input/output	2–3 months	Wider accessibility
Phase 4	Cloud deployment and progress analytics	3–4 months	Public access and insights